reesabp 's 2014 User-Submitted Test - Repaired practice edition - Answer key

Division B | Heredity | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

ANSWER KEY - Review only after submitting.

q67. (6 points) Authored replacement for source item 67: Solve this case and explain your reasoning: A diploid cell with 2n = 8 undergoes meiosis. How many chromosomes and chromatids are in each cell at the end of meiosis I and at the end of meiosis II?

reasoning: Before meiosis, replication gives 8 chromosomes with 16 chromatids.
Meiosis I separates homologs: each cell has 4 chromosomes, still with 2 chromatids each (8 chromatids).
Meiosis II separates sister chromatids: each of four cells has 4 chromosomes with 1 chromatid each. (3 points).

conclusion: After meiosis I: 4 chromosomes and 8 chromatids per cell. After meiosis II: 4 chromosomes and 4 chromatids per cell. (3 points).

repair-case. (4 points) Authored written substitute for unavailable source material: Name an experimental question that PCR could answer.

Unit: PCR and Sanger sequencing.
Worked-case data: A PCR starts with 50 copies of a target. Assuming perfect doubling, how many copies are present after 10 cycles?
Separate investigation observations:
Positive control: Known target DNA gives a 450 bp band.
Negative control: Water gives no band.
Sample 1: One band at 450 bp.
Sample 2: No band, but a later test shows its DNA is intact.

answer: Whether a sample contains DNA from a particular pathogen, detected by whether a target-specific product forms. (4 points).